Panskura Banamali College (Autonomous)

M.Sc. 2nd Semester Examination - 2021 (Online Mode)

Subject: Computer Science **Paper: COS-201**

Paper Name: Advanced Database Management System

F.M.: 40 **Time : 2 Hours**

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1. Answer any two questions: **(2 x 4 = 8)**

a. What do you mean by locking techniques of concurrency control?
b. What do you mean by query and sub query?
c. What are the various characteristics of SQL?

2. Answer any two questions: **(2 x 8 = 16)**

a. Write a short note on '*normalization*'. Write the different database anomalies.
b. Define multi-valued dependency? Write the concept of serializable schedule?
c. What is ACID property of DBMS? Explain.

3. Answer any one question: **(1x 16 =16)**

a. What is read before write protocol problem? Explain with example. What is query optimization? Write a short note on DCL. (5+5+6)
b. What is differed database modification technique? What is hierarchical database model? What is database deadlock? How deadlocks are detected and recovery is performed? Explain. (4+5+7)

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